

MASS-APP-014-V10 — Knowledge Discovery & Enterprise Search

Document Control

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1. Purpose

V10 provides explainable, context-aware discovery across governed organizational knowledge. It retrieves references from authoritative systems without becoming their system of record, changing source content, or bypassing source authorization.

2. Architectural Boundary

V10 owns discovery sessions, query interpretation, source selection, result assembly, ranking evidence, citations, saved searches, and user feedback. ENG-007 owns enterprise knowledge; ENG-009 owns AI orchestration; ENG-015 governs APIs; ENG-024 owns enterprise analytics; APP-013 owns design artifacts. V10 neither copies these responsibilities nor performs autonomous action.

3. Platform Consumption Map

Consumes	Use
ENG-003 / ENG-004	Identity, tenant context, authorization
ENG-005	Discovery lifecycle events
ENG-007	Governed knowledge retrieval
ENG-009	Query interpretation and synthesis
ENG-011	Audit and observability
ENG-015	API contracts
ENG-024	Approved analytical references
ENG-027	Information lineage and ontology context
APP-013	Read-only artifact references
APP-014 V01–V04, V08–V09	Sessions, memory, reasoning, agents, communication and execution context

Published events: discovery.started, discovery.completed, discovery.failed, saved-search.created, search-feedback.recorded.
Consumed events invalidate stale source snapshots without modifying the source.

4. Role Mapping

V10 role | Baseline mapping | Authority

Search Viewer | Viewer specialization | Run searches and inspect authorized results

Search Contributor | Contributor specialization | Save searches and submit relevance feedback

Discovery Steward | Steward specialization | Govern source profiles, ranking policies, and review quality

Discovery Administrator | Administrator specialization | Configure tenant discovery controls and audit access

These roles extend through specialization; they do not replace APP-014 baseline roles.

5. Capability Model

Query Intake

Accepts natural-language or structured queries with tenant, user, filters, and optional session context. Outputs an immutable query record and correlation identifier.

Source Resolution

Selects only registered, active sources the principal may access. Every result remains linked to its source type, source identifier, version, and observed timestamp.

Retrieval and Ranking

Combines lexical, metadata, and governed semantic signals. Ranking records preserve signal contributions so ordering can be explained. V1 supports configurable weighted ranking; future models may improve ranking without changing result contracts.

Synthesis and Citation

ENG-009 may summarize retrieved evidence. Each synthesized statement must retain citations to accessible source records. Unsupported conclusions are marked uncertain rather than presented as fact.

Feedback and Saved Discovery

Users may save query definitions and record relevance feedback. Feedback informs tenant-local tuning only and never creates cross-tenant learning.

6. Discovery Lifecycle

Received → Authorized → Resolved → Retrieved → Ranked → Synthesized → Delivered → Preserved

Failures terminate with a governed error and audit event. Partial results identify unavailable sources and never silently imply completeness.

7. Data Model

Entity | Responsibility

DiscoverySession | Tenant-scoped discovery context

SearchQuery | Immutable submitted query
SearchSource | Governed source registration
QuerySource | Sources selected for a query
SearchResult | Ranked source reference
ResultCitation | Evidence location and lineage
RankingExplanation | Signal contribution record
SavedSearch | Reusable query definition
SearchFeedback | User relevance assessment
SearchAudit | Security-relevant discovery history

All tenant-owned foreign keys use tenant-safe composite relationships. Polymorphic source identifiers require application validation against the registered source adapter and current authorization.

8. API Contracts

Method | Path | Purpose
POST | /discovery/search | Execute governed discovery
GET | /discovery/queries/{id} | Retrieve query and status
GET | /discovery/queries/{id}/results | Retrieve ranked results
GET | /discovery/results/{id}/citations | Retrieve evidence citations
GET | /discovery/sources | List authorized source profiles
POST | /saved-searches | Save a query definition
GET | /saved-searches | Browse saved searches
PATCH | /saved-searches/{id} | Update owned saved search
DELETE | /saved-searches/{id} | Soft-delete saved search
POST | /discovery/results/{id}/feedback | Record relevance feedback
GET | /discovery/audit/{queryId} | Steward audit view

Search submission accepts an idempotency key. Repeated requests return the original query. Forbidden sources are omitted and recorded in the audit without disclosing their content.

9. Security and Integrity

- RLS applies to every tenant-owned table.
- Source authorization is evaluated at query time and result-read time.
- Search results contain references and permitted excerpts, not an uncontrolled shadow repository.
- Citations are immutable after query completion.
- Feedback cannot alter source truth or ranking history.
- Deleted or archived sources remain historically cited but are unavailable for fresh retrieval.

Implementation-grade definitions are in MASS-APP-014-V10_Migration_Reference.sql.

10. Failure Behavior

Unavailable sources yield explicit partial-completion metadata. Failed synthesis returns ranked evidence without synthesis. Authorization changes suppress inaccessible results on subsequent reads. Retry applies only to safe retrieval operations and does not duplicate queries.

11. V1 and Future Evolution

V1 implements tenant-scoped source registration, governed query execution, deterministic ranking configuration, citation-preserving synthesis, saved searches, and feedback. Future ranking refinement, richer federated discovery, and additional adapters attach through existing source and ranking contracts; they are not V1 requirements.

12. Constitutional Boundary Statement

V10 discovers and explains authorized enterprise knowledge. It does not own knowledge, documents, analytics, relationships, decisions, workflows, communication delivery, or source-system mutation. Human users retain interpretive and decision authority.